Geodetic inversion using the VSM package





https://github.com/EliTras/VSM_test

VICTOR

Volcanic and Seismic source Modelling (VSM) is an open source Python tool to model ground deformation detected by satellite and terrestrial geodetic techniques.

Forward analytical models available and freely combinable

- isotropic point-source
- finite volume sphere
- penny-shaped crack
- spheroid
- moment tensor
- rectangular dislocation

Input data supported

- interferometric SAR
- GNSS
- levelling
- electronic distance measurements
- tilt
- strain

Volcanic and Seismic source Modelling

VICTOR

Sampling algorithms

- global optimization algorithm based on the Voronoi cells (Neighbourhood Algorithm - NA)
- probabilistic approach to parameters estimation based on the Bayes theorem (Bayesian Inference - BI)

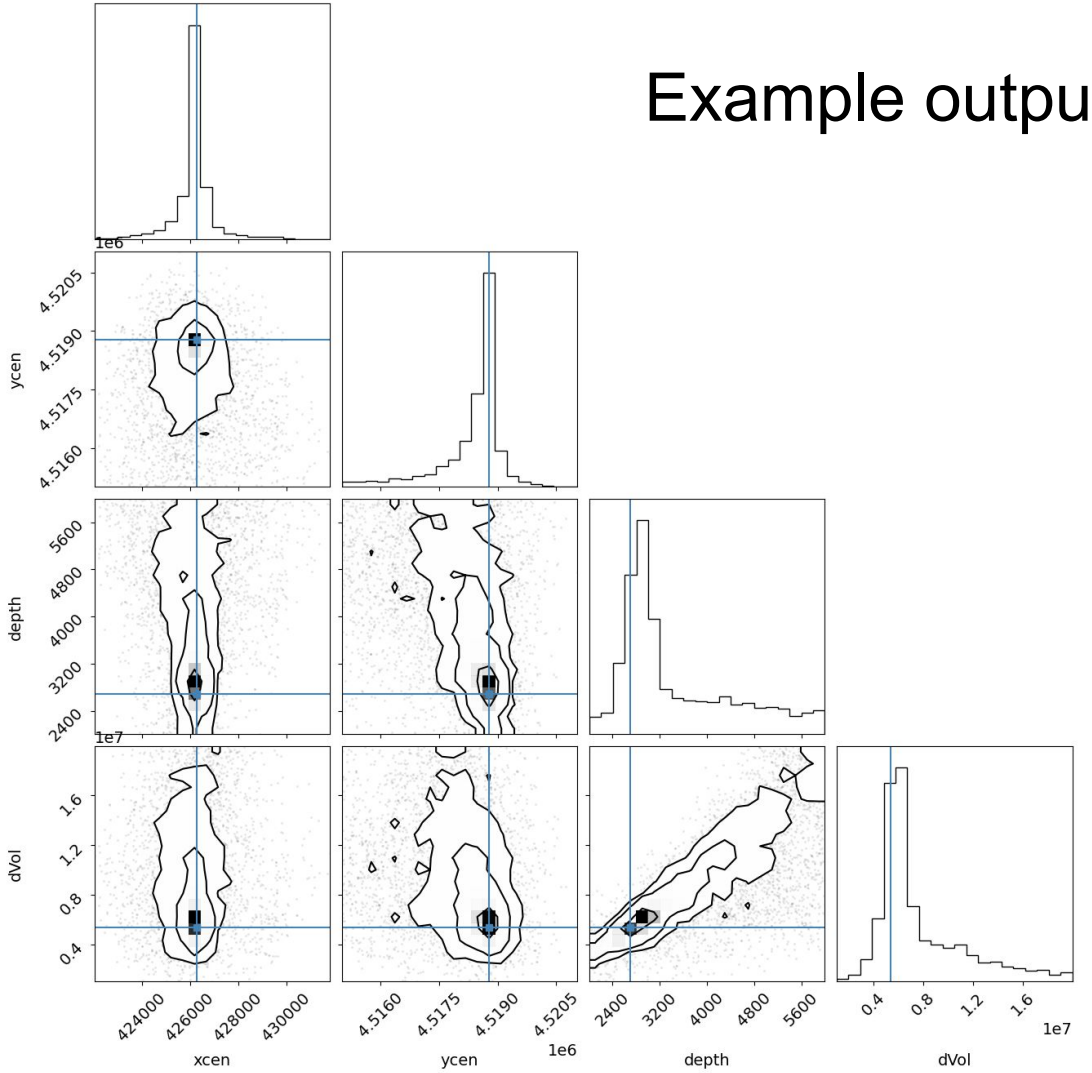
VSM can be executed as Python script, in Jupyter Notebook environments or by its Graphical User Interface.
Tests available at https://github.com/EliTras/VSM_test

For any inquires, please write to elisa.trasatti@ingv.it
Trasatti, E. (2022). "Volcanic and Seismic source Modelling (VSM). An Open Tool for Geodetic Data Modeling", Frontiers in Earth Science, doi: 10.3389/feart.2022.917222

Activity instructions

- Copy the folder `~/shared/Education/2025\ Idaho\ Workshop/VSM/` to your home directory
- Go to the Sill example: folder VSM → USECASE → Sill2NA, and double-click the notebook `VSM_Sill2NA.ipynb`
- Step through the Notebook's cells and explore
- Repeat for the notebook in folder Sill2BI
- To run other cases, go to VSM folder, and double-click the VSM_utilities notebook `VSM_utilities.ipynb`
- Set the test case folder and input file name that you would like to run, and run the notebook.

Example output from VSM (Mogi source)



Probability distributions for:

- X, Y, and Depth of the center
- the change in volume

Example output from VSM (Mogi source)

